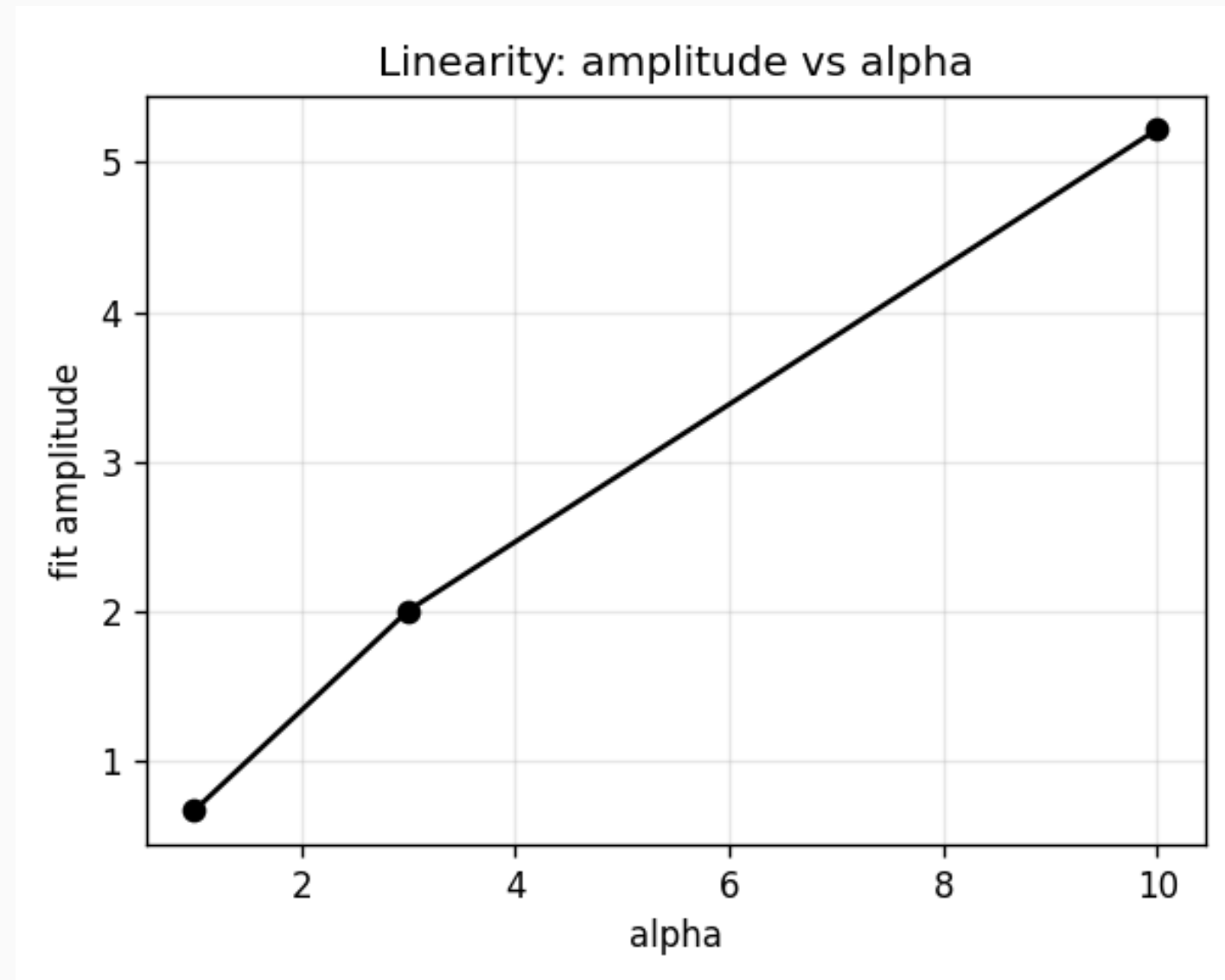


3. Linearity + phase stability

Two-pillar causal test



- Amplitude grows **linearly** with α ($R^2 \geq 0.993$).
- Phase drift $\leq 3^\circ$ across $\alpha \in \{1, 3, 10\}$.
- Random planes and permuted labels kill both signals — so the plane is not a geometric accident.